

Rudransh Srivastava

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Summary: A dynamic tech enthusiast from Varanasi, India, I am passionate about innovation, leadership, and computer science. My achievements include developing impactful projects and winning accolades in national & international tech competitions. As the school's Head Boy, I've established thriving clubs and organized events for students throughout high school. I approach challenges with technical expertise, leadership, and a commitment to creating solutions that make a real difference.

Education:

Sunbeam English School, Varanasi, Uttar Pradesh

Mar 2021 – Apr 2025

High School - CBSE, 12th: Physics, Mathematics, Chemistry, Computer Science, Entrepreneurship & English
SAT: 1370 (750 math; 620 eng)

Work Experience:

Think Startup Youth Ideathon

Aug 2023-Nov 2023 | Jun 2024-Nov 2024

Project Evaluator'23 & Executive Intern'24

Supported India's largest ideation festival by evaluating project presentations, supervising evaluators and organizing team members, mentoring participants, and recruiting campus ambassadors.

- Assessed over 60 project presentations at the ThinkStartup Youth Ideathon based on innovation, feasibility, social impact, and presentation quality.
- Organized feedback sessions and provided constructive feedback to participants to enhance their prototypes and improve presentation skills, ultimately helping three teams reach the podium.
- Recruited and supervised 25+ campus ambassadors across India, ensuring effective representation and outreach for students.

Luzeblaze

Dec 2023 - Feb 2024

Full-stack Web Development Intern

Assisted in API and frontend development by writing secure data exchange endpoints and creating responsive user interfaces, ensuring seamless functionality across various devices.

- Developed RESTful APIs using Express.js, optimizing server performance and implementing secure, efficient data handling with robust, asynchronous endpoints to enhance scalability and efficiency.
- Developed dynamic, responsive user interfaces using Next.js and Tailwind CSS, enhancing cross-platform compatibility and user engagement.

Hack Club, SBS BGN

Jun 2022 - Feb 2025

Co-founder'22 & President'23

Co-founded and pioneered the city's inaugural HackClub, establishing one of the earliest state-level chapters to promote coding and innovation among students.

- Reached over 1,000 students through more than 10 affiliated branches and independent organizations, fostering a collaborative learning environment.
- Developed over 10 in-house projects, securing \$5,000 in prizes and grants across 10+ science fairs and hackathons.

Stellar Scholar, STEM Club

Mar 2024 - Feb 2025

Co-founder & CTO

Co-founded a student-led organization dedicated to orchestrating STEM events and research for students.

- Developed a platform to empower high school students to engage in STEM projects, conduct research and write papers, and receive expert guidance to advance their scientific and technological pursuits.
- Organized and managed weekly STEM events, including hackathons, coding workshops, and innovative idea pitch sessions, published tech-related newsletters, and managed a community of 500+ people. Also, developed and managed the club website.

The Ennead International MUN @ Sunbeam Bhagwanpur
Head of Technology

Jul 2022 - Aug 2022

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Led and supervised the technical team for the Ennead International MUN held at Sunbeam English School with over 100 delegates.

- Supervised the creation of Banners, Standees, Flyers, Motion Graphics, and Promo videos for the event.
- Developed and managed the MUN website, and implemented registration and login functionalities for streamlining the registration process.
- Provided technical support to over 100 delegates, managing breakout rooms and resolving technical issues efficiently during the event.

Student Council, Sunbeam Bhagwanpur

Jun 2023 - Mar 2024

Head Boy

- Served as Head Boy, leading the planning and execution of major school fests and events, while working closely with faculty on student welfare and institutional initiatives.
- Founded and structured 5+ student clubs to expand co-curricular participation, foster diverse interests, and uphold discipline and student engagement across the school community.

Projects:

- **Mini-Vent:** an economical and portable medical ventilator meant to aid in emergencies.
 - Engineered a cost-effective and portable emergency ventilation system, by utilizing a high-torque DC motor to create an oscillating compression mechanism that compresses an Ambu airbag, ensuring essential respiratory support for patients in critical conditions.
 - Implemented an array of sensors, including a pulse oximeter for measuring blood oxygen levels and ECG and EMG modules for monitoring the electrical activities of the heart and muscles, all controlled by a microcontroller to provide real-time vital sign monitoring.
 - Developed a cross-platform mobile and web application that leverages an API to facilitate remote patient monitoring. The system securely stores and organizes patient vitals in a MongoDB database, enabling healthcare professionals to access real-time data and make timely decisions.
- **Eye-Blink:** a real-time drowsiness detection system aimed at enhancing road safety.
 - Used Python OpenCV for facial feature extraction and analysis. The system performs a comprehensive assessment of critical facial attributes, including eye movement dynamics, blink pattern recognition, and precise head pose tracking.
 - Integrated infrared cameras to improve detection in low-light conditions and employed a Raspberry Pi for efficient processing and deployment.
 - Integrated the drowsiness detection system with the vehicle's computer-inhibited ABS brake system to engage the brakes if drowsiness is detected automatically.
- **TubeAI:** an innovative YouTube summarization bot equipped with intelligent chatbot functionality.
 - Uses OpenAI's Whisper model and GPT3.5 to generate accurate content summaries and respond to user queries, regardless of video caption availability.
 - Uses a two-step approach: extracting captions via the YouTube-transcript API when available, and employing Whisper for audio transcription when captions are absent.
 - Implemented a chatbot that provides interactive and accurate responses to user queries about YouTube video content.

Achievements:

- Winner: Inspire Awards MANAK, Intel & DST, India, selected among 10 lakh+ nominations.
- Winner: Think Startup Youth Ideathon 2022, held at IIT Delhi, selected among 90,000+ participants.
- Runner-Up: Unfold 2023, Coin DCX multi-chain hackathon @ KTPO Bangalore with 1,000+ participants.
- Winner: HACK JKLU, hackathon @ JK Lakshmipat University with 1,000+ participants.
- 3rd in TCL B-Plan Competition, Think Startup @ Prometheus School out of 3,500+ participants.
- Winner: ATL Marathon 2023-24, Niti Aayog, GoI, placing in the top 500 out of over 20,000 entries.
- Finalist: Vivo Ignite 2023 – Top 200 Teams out of 19,000+ teams.
- Winner: Innovate4Impact Smart City Challenge, La Trobe University, Australia.
- 1st in Teenovations: Robotics model making with computer scraps @ COFAS, CMS Lucknow.

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- Publication: Mini-Vent - An Economical Miniaturized Intensive Care Unit System, IRJET.
- Best Intern Award by Luneblaze: surpassed over 200 interns in full-stack web development.